

CONSTRUCTION OS

PRD - Approvals & Workflow Engine

Implementation-ready Product Requirements Document for Product, UX/UI, Frontend, Backend, Platform, Security, QA, DevOps and Business Operations Teams

Product purpose

Provide a versioned, permission-aware and auditable decision-orchestration engine that applies controlled approval, review, verification, acknowledgement, escalation, delegation and finalization processes to records owned by every ConstructionOS module.

Document control	Value
Document	PRD_Approvals_Workflow_Engine
Version	1.0
Status	Implementation-ready functional and technical specification
Audience	Product, UX/UI, frontend, backend, platform, security, QA, DevOps, finance, legal, HR, procurement and project operations teams
Primary surfaces	Internal web, responsive mobile, native mobile approval views, external portal workflow projections and administration console
Core architecture	Versioned workflow definitions + durable runtime + approval matrices + dynamic participant resolution + source-module action coordination
Access model	Deny-by-default, tenant/company/project/record aware, separation-of-duty controlled and fully auditable
Source-of-truth policy	Operational modules own business records and consequences; Workflow owns routing, work items, decisions, delegation, SLA and decision history
Initial integration policy	Self-contained engine with first-party integrations to ConstructionOS modules. External e-signature and identity providers remain roadmap items.
Platform Configuration	Mandatory for every workflow, stage, node, participant resolver, matrix, decision, lock, SLA, escalation, override and source action
Date	2026-08-06

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1. Executive Summary

ConstructionOS requires one platform-wide mechanism for controlled decisions. Without a central engine, each module would hard-code its own approval chain, delegation behavior, timers, lock rules and decision history. That would create inconsistent governance, duplicated implementation, security gaps and difficult audits.

The Approvals & Workflow Engine solves this by separating business records from decision orchestration. A source module submits an exact record version. The engine selects an applicable workflow, resolves participants, creates durable work items, records immutable decisions, handles time and escalation, and asks the source module to apply the approved business transition. The workflow is complete only after the source module confirms finalization.

Permanent architectural rule

Source modules own what is being decided and the resulting business consequences. The Workflow Engine owns how the decision is routed, performed, delegated, escalated, evidenced and audited.

The initial release must support sequential and parallel decisions, value-based approval matrices, dynamic approver resolution, separation of duties, information requests, return for correction, record locking, version invalidation, durable SLA timers, delegation, escalation, mobile approvals, external portal participation and controlled emergency exceptions.

2. Product Vision and Business Outcomes

The product vision is to make every important decision in ConstructionOS consistent, explainable, timely and reconstructable. Users should have one approval inbox, one decision experience and one trusted history, regardless of whether the underlying item is a purchase order, contract, payroll run, legal filing, HSE release or client variation.

- **Governance consistency** - replace module-specific approval code with a reusable platform capability.
- **Operational speed** - route requests immediately to the correct authorized people and surface bottlenecks.
- **Auditability** - preserve the exact submitted version, decision evidence, participant resolution and source finalization outcome.
- **Security** - prevent self-approval, unauthorized delegation, hidden bypasses and access leakage.
- **Configurability** - allow companies and projects to apply different workflows without code changes.
- **Scalability** - support high-volume financial approvals and long-running legal or project processes.
- **Implementation reuse** - give all modules a standard trigger, lock, decision and completion contract.

3. Success Metrics and Guardrails

Metric	Initial target	Guardrail
Workflow selection success	>= 99.9% of valid submissions select one deterministic active workflow	No hidden fallback; every fallback is explicit and audited
Decision processing	P95 under 3 seconds excluding source-module finalization	No optimistic final approval before server confirmation
Approval queue load	P95 under 2 seconds for normal tenant workloads	Permission checks remain enforced
SLA timer accuracy	No missed durable timer after restart or deployment	Business calendar and pause history preserved
Duplicate prevention	Zero duplicate workflow instances for the same idempotency key	Retries return the original result
Audit completeness	100% of decisions include identity, record version, workflow version and timestamp	No editable finalized decision
Source finalization reliability	>= 99.5% automated completion; failures visible and retryable	Workflow stays pending until source confirmation
Access isolation	Zero cross-tenant or unauthorized cross-project exposure	Aggregation and notifications cannot infer restricted data
Adoption	At least 90% of configured controlled actions use the engine after rollout	No forced migration before a source module is ready

Guardrails: workflow analytics must not automatically become employee-performance ratings; platform administrators must not gain business approval authority; and emergency override must never erase skipped stages or previous decisions.

4. Scope, Non-Goals and Assumptions

In scope

- Workflow definitions, immutable versions and controlled publication.
- Human review, approval, verification and acknowledgement work items.
- Sequential, parallel, quorum, majority and required-participant routing.

- Dynamic participant resolution, approval matrices, delegation and substitutes.
- Separation-of-duty and conflict-of-interest controls.
- Information requests, return for correction, resubmission, rejection and cancellation.
- Record locks, material-change policies and approval invalidation.
- Durable SLA timers, business calendars, reminders and escalation.
- Automated source validations and source-module action requests.
- Mobile decision experience and controlled external portal stages.
- Emergency override, configured bypass, simulation, migration, analytics and audit.

Non-goals

- Replacing Tasks as the platform execution system.
- Replacing Documents, Contracts, Finance or other source modules.
- Providing unrestricted arbitrary-code automation.
- Acting as a general-purpose chat or messaging product.
- Equating an internal platform approval with a legally qualified electronic signature.
- Allowing offline approval finalization without server validation.
- Providing third-party e-signature, identity or government filing integrations in the initial release.

Assumptions

- Core user, company, project, permission and audit services already exist.
- Source modules expose versioned records and controlled state-transition APIs.
- Notifications & Event Centre handles delivery channels and message preferences.
- Documents owns files and document revisions referenced by workflows.
- Platform Configuration is available to define tenant and company-specific behavior.

5. Terminology and Core Concepts

Term	Definition
Workflow Definition	Configurable design of a decision process associated with one or more source entity types.
Workflow Version	Immutable published snapshot of a definition. Active instances retain their original version.
Workflow Instance	Durable execution for one source record and one submitted source-record version.
Stage	Business grouping of one or more workflow nodes, such as Technical Review or Executive Approval.
Node	Executable unit: human decision, condition, parallel split/join, timer, validation, source action or end.
Work Item	A decision request assigned to a user, role or resolved participant group.
Decision	Immutable response to a work item, including identity, result, time, evidence and source version.
Approval Matrix	Effective-dated routing table that maps business dimensions and thresholds to required participants.
Source Action	Version-aware request for the owning module to apply a status transition or business consequence.
Override	Exceptional temporary authority that bypasses or modifies normal routing and requires post-event review.
Bypass	Configured exemption proving that a workflow or stage is not applicable under a known rule.

6. Source-of-Truth Boundaries

Capability	Workflow Engine owns	Source module owns
Record content	Submitted snapshot reference and decision-visible projection	Authoritative fields, documents and business validation
Status	Workflow runtime status	Business record status
Authority	Routing, participants, work items and decisions	Who may initiate the business transition
Locking	Lock instruction and lock metadata	Actual enforcement on the record and fields

Capability	Workflow Engine owns	Source module owns
Completion	Decision completion and source-action request	Final state transition and business side effects
Audit	Workflow execution and decision trace	Record-level change history and business audit
Notifications	Event publication	No direct channel delivery; Notification Centre delivers
Source submission -> Workflow selection -> Work items -> Decisions -> Source finalization request -> Source confirmation -> Workflow completed		

Approval and completion are deliberately separate. A workflow can be approved but remain in Source Finalization Pending when the source module reports a validation error, version conflict or temporary outage.

7. Personas and Stakeholders

Persona	Primary needs	Restrictions
Requester	Submit, track, answer questions, correct and resubmit	Cannot approve own request where separation rules apply
Approver	Review exact version, understand impact, decide quickly	Only configured scopes, capabilities and values
Reviewer / Verifier	Provide advisory or mandatory specialist response	May not have final approval authority
Manager	See team queue, bottlenecks and escalations	Cannot see confidential content outside granted scope
Workflow Administrator	Design, test, publish and monitor workflows	Does not gain live business approval authority
Platform Administrator	Configure engine availability and system controls	No automatic access to confidential records or decisions
Auditor / Compliance	Reconstruct selection, routing, decisions and overrides	Read-only and purpose-limited
External Portal Participant	Submit or respond in controlled external stages	No internal workflow builder or internal-only history
Mobile Field Approver	Review and decide from mobile	Online confirmation required for final decision

8. Product Surfaces and Navigation

Primary navigation

- My Approvals
- My Submissions
- Delegated Approvals
- Team Approval Queue
- Escalated Items
- Completed Decisions
- Workflow Dashboard
- Workflow Definitions
- Approval Matrices
- Workflow Builder
- Workflow Versions
- Workflow Instances
- Delegations
- SLA and Escalation Policies
- Workflow Simulation
- Workflow Analytics
- Failed Workflows
- Administration and Settings

Embedded surfaces

Every source module may embed a workflow status card, current stage, pending participants, decision history, lock state and Submit / Withdraw / Resubmit actions. The embedded view must never duplicate the authoritative workflow runtime.

9. Functional Requirements Register

WF-FR-001 - Deterministic workflow selection: Select exactly one active workflow version for a valid trigger using auditable rules.

Acceptance: Given identical inputs and configuration versions, the same workflow is selected.

WF-FR-002 - Idempotent start: Prevent duplicate workflow instances for retried or repeated triggers.

Acceptance: Reusing an idempotency key returns the original instance.

WF-FR-003 - Version binding: Bind every instance and decision to an exact source-record version.

Acceptance: The approval detail identifies the version and material changes invalidate per policy.

WF-FR-004 - Sequential routing: Support ordered decision stages.

Acceptance: A downstream stage cannot begin before the configured upstream completion condition.

WF-FR-005 - Parallel routing: Support multiple concurrent branches and configurable join rules.

Acceptance: Join waits for all, quorum, majority or required-role conditions as configured.

WF-FR-006 - Dynamic participants: Resolve approvers from roles, records, matrices, organization and relationships.

Acceptance: Resolution trace shows included and excluded candidates.

WF-FR-007 - Separation of duties: Block or reroute prohibited self-approval and conflicting duties.

Acceptance: A violation produces a visible resolution result and audit event.

WF-FR-008 - Immutable decisions: Prevent edits to finalized decisions.

Acceptance: Correction creates a new auditable action, never an in-place mutation.

WF-FR-009 - Information request: Pause or continue SLA according to policy while information is requested.

Acceptance: Question, response, attachments and timing are preserved.

WF-FR-010 - Return and resubmit: Route a record to a configured correction point and validate resubmission.

Acceptance: Previous decisions are preserved or invalidated according to explicit rules.

WF-FR-011 - Record locking: Coordinate field, document or full-record locks with the source module.

Acceptance: The engine records the source confirmation before treating a lock as active.

WF-FR-012 - Material change handling: Detect submitted record changes and apply configured invalidation.

Acceptance: Changed fields and impacted stages are visible.

WF-FR-013 - Delegation: Allow bounded, effective-dated delegation without exceeding authority.

Acceptance: Decision records both delegator and acting user.

WF-FR-014 - SLA and escalation: Calculate durable timers using configured calendars and pause rules.

Acceptance: Restart or deployment does not lose timer state.

WF-FR-015 - Source finalization: Request and confirm the source-module transition after workflow approval.

Acceptance: Workflow does not show Completed until source confirms.

WF-FR-016 - Mobile decisions: Provide mobile review and online-confirmed decision actions.

Acceptance: Final state is shown only after server acceptance.

WF-FR-017 - Portal stages: Support limited external submission, response and acknowledgement nodes.

Acceptance: External users see only published external status and content.

WF-FR-018 - Simulation: Allow dry-run testing without live records or decisions.

Acceptance: Simulation outputs route, participants, SLA and errors.

WF-FR-019 - Emergency override: Support controlled exceptional authority with mandatory post-event review.

Acceptance: Skipped stages remain visible and cannot be erased.

WF-FR-020 - Audit completeness: Preserve the complete selection, execution, decision and finalization trace.

Acceptance: Auditor can reconstruct the workflow without relying on application logs.

10. Workflow Dashboard

The dashboard provides operational control rather than only reporting. Every number opens the underlying workflow instances and respects record permissions.

- Awaiting decision, overdue, escalated, returned, rejected and source-finalization-failed counts.
- Approval volume and duration by module, company, project and workflow definition.
- High-value and high-risk items requiring attention.
- Unresolved participants, inactive approvers and delegation conflicts.
- SLA compliance, bottleneck stages and queue age.
- Emergency overrides and bypasses pending review.
- Failed integrations, dead-letter events and retryable source actions.

WF-FR-021 - Dashboard drill-through: Every dashboard metric must open a permission-filtered list with the exact same metric definition.

Acceptance: Counts reconcile with the resulting list and expose refresh time.

11. My Approvals and Team Queues

My Approvals

- Pending
- Due soon
- Overdue
- Information requested
- Delegated to me
- Escalated to me
- Decided recently
- Saved filters and views

Team Approval Queue

Managers with the correct capability may view queue health, not automatically the full confidential source content. Queue cards show safe metadata, source type, project/company, stage, age, SLA, priority and assignment status.

WF-FR-022 - Queue filtering: Support module, company, project, workflow, stage, priority, SLA, value band, requester and confidentiality filters.

Acceptance: Filters are URL-addressable and preserve permission boundaries.

WF-FR-023 - Bulk actions: Permit bulk acknowledgement or low-risk configured decisions only where all items share the same allowed action and context.

Acceptance: Bulk processing produces one immutable decision per work item and reports partial failures.

12. Approval Detail Page

1. Decision summary
2. Source-record overview
3. Important values and risk indicators
4. Changes since prior submitted version
5. Controlled documents
6. Previous decisions
7. Workflow timeline
8. Information requests and comments
9. Decision conditions
10. Decision panel
11. Audit context

Decision clarity requirement

The page must tell the approver exactly what is being decided, which version is under review, why this user received the item, what happens after each decision, and whether any other stages remain.

WF-FR-024 - Version comparison: Show material field and document changes against the previous submitted or approved version.

Acceptance: Approver can expand each changed value and open the source record where permitted.

WF-FR-025 - Decision validation: Before accepting a decision, revalidate user, work-item state, source version, required comment, evidence and separation rules.

Acceptance: Stale or unauthorized decisions are rejected without losing the drafted comment.

13. Workflow Definition Model

Field group	Required content
Identity	Workflow key, name, description, category, owner and source entity types
Applicability	Tenant, company, project, department, subtype, value, risk and effective dates
Runtime	Priority, default SLA, cancellation policy, change policy and completion action

Field group	Required content
Security	Confidentiality, participant capabilities, field projection and export restrictions
Version	Version number, status, created by, published by, effective date and superseded version
Failure	No-match behavior, participant failure, source-action failure and administrative intervention policy
Draft -> Testing -> Published -> Active -> Superseded -> Deprecated -> Archived	

WF-FR-026 - Immutable publication: A published version cannot be edited.

Acceptance: Any change creates a new draft version with an explicit comparison to its parent.

14. Visual Workflow Builder

The builder is a node-based administration surface with a canvas, node library, properties panel, validation panel, version history and simulation tools.

Node type	Purpose
Start	Receives trigger and input context
Human Approval	Creates a formal decision work item
Human Review	Creates mandatory or advisory specialist review
Verification	Confirms evidence or source condition
Acknowledgement	Records receipt or understanding
Condition	Evaluates a governed expression
Parallel Split / Join	Starts and reconciles concurrent branches
Timer / Escalation	Schedules durable reminders or rerouting
Automated Validation	Calls a source-owned validation
Source Action	Requests a controlled business transition
External Response	Waits for an authorized portal participant
End / Cancellation / Override	Completes or terminates a defined path

WF-FR-027 - Builder validation: Detect missing start/end, unreachable nodes, invalid conditions, circular paths, missing joins, empty participant rules and unsupported source actions.

Acceptance: Publication is blocked while any error-level validation remains.

15. Triggers and Workflow Selection

Supported triggers

- User submission
- Source domain event
- Scheduled compliance or renewal review
- Authenticated internal API call

Selection dimensions

- Tenant and legal company
- Project and department
- Module, entity and subtype
- Currency-normalized value
- Risk, severity and confidentiality
- Location, client, supplier, funding source and custom fields

WF-FR-028 - Selection trace: Store evaluated rules, inputs, selected version and fallback reason.

Acceptance: Administrators can reproduce selection using the historical configuration versions.

WF-FR-029 - No-match handling: Support block, explicit default, administrator triage or configured no-approval exemption.

Acceptance: No implicit first-match behavior is allowed.

16. Approval Matrices

Purchase Order value	Required route
Up to EUR 5,000	Project Manager
EUR 5,001 - 25,000	Project Manager + Procurement Manager
EUR 25,001 - 100,000	Procurement Manager + Finance Director
Above EUR 100,000	Finance Director + CEO

Matrices may also use company, project, department, entity type, risk, category, cost centre, supplier type, funding source and custom governed dimensions.

WF-FR-030 - Effective-dated matrices: Prevent overlapping active ranges for the same dimension set and preserve historical versions.

Acceptance: Publication validation reports gaps, overlaps and missing fallback routes.

WF-FR-031 - Currency normalization: Store the rate, source, date and normalized amount used for threshold evaluation.

Acceptance: Historical route remains reproducible after exchange rates change.

17. Participant Resolution

- Explicit user
- Record responsible user
- Line manager or manager-of-manager
- Project or site role
- Department or company role
- Finance, legal, HSE, QA/QC, HR or procurement role
- Budget or cost-centre owner
- Approval matrix result
- External portal contact
- Custom governed resolver

WF-FR-032 - Participant eligibility: Verify active account, scope, capability, conflict rules, delegation and access to decision content.

Acceptance: Resolution records all candidates and the reason each was included or excluded.

WF-FR-033 - Unresolved participant: Do not silently skip a mandatory participant.

Acceptance: Instance enters a visible intervention state and follows the configured fallback or escalation path.

18. Separation of Duties and Conflict Rules

- Requester cannot approve own request where configured.
- Creator cannot be final approver for financial or procurement records.
- Payment preparer cannot be sole payment approver.
- Payroll calculator cannot be sole payroll approver.
- A supplier evaluator with a declared conflict is excluded.
- Emergency override initiator cannot independently close the post-event review.

WF-FR-034 - Conflict resolution: Block, exclude, substitute, escalate or require an additional approver according to policy.

Acceptance: The system never resolves a violation by silently granting authority.

WF-FR-035 - Conflict declaration: Allow authorized users to declare a conflict before deciding.

Acceptance: The work item is paused and rerouted without exposing unnecessary reason details.

19. Sequential Routing

Requester -> Department Manager -> Finance Review -> Director Approval -> Source Finalization

Each stage defines entry conditions, participant rules, required decisions, rejection behavior, return destination, SLA and completion transition.

WF-FR-036 - Sequential gating: Do not activate downstream stages until upstream completion is durably committed.

Acceptance: Concurrent retries cannot create duplicate downstream work items.

20. Parallel Routing, Quorum and Voting

Technical Review --\ Financial Review ----> Join -> Final Decision Legal Review -----/	
Rule	Behavior
All approve	Every required branch must approve
Any one approves	First valid approval completes the join
Majority	Approvals must exceed rejections among eligible voters
Quorum	Minimum responses and approvals required
Required roles	Named required roles must approve, optional roles remain advisory
Weighted vote	Configured weights determine result; reserved for governance use cases

WF-FR-037 - Concurrent decisions: Handle simultaneous branch decisions with optimistic concurrency and deterministic join evaluation.

Acceptance: Exactly one transition is committed when completion criteria are first met.

21. Decision Model

Decision option	Typical effect
Approve	Contributes to successful stage completion
Reject	Follows configured termination or correction path
Return for correction	Routes to a named source or workflow stage
Request information	Pauses or continues the work item while requesting a response
Approve with conditions	Creates enforceable follow-up conditions
Acknowledge	Confirms receipt or understanding, not business authorization
Abstain	Counts according to quorum policy
Not applicable	Valid only where the workflow explicitly allows it

WF-FR-038 - Decision immutability: Finalized decisions cannot be edited or deleted.

Acceptance: Withdrawal, reopening or correction produces a new linked record and preserves the original.

WF-FR-039 - Decision evidence: Support mandatory reason codes, comments, attachments and reauthentication by decision type.

Acceptance: The UI blocks submission until all configured evidence is present.

22. Conditional Approval

Approve with conditions is a first-class decision, not a comment. Each condition includes description, responsible party, due date, evidence requirement, verification step, and effect on source execution.

- Approved but blocked from execution
- Provisionally approved
- Approved with follow-up obligation
- Approved only after evidence verification

WF-FR-040 - Condition tracking: Create linked condition records and prevent final source transition where the policy requires closure first.

Acceptance: Conditions remain visible in the workflow and source-module projection.

23. Information Requests, Returns and Resubmission

Decision Pending -> Information Requested -> Requester Responding -> Information Submitted -> Decision Resumed

Information requests preserve the exact question, requester, response, attachments, due date, count and SLA pause behavior. Return for correction additionally identifies the destination stage, required corrections and whether earlier decisions remain valid.

WF-FR-041 - Resubmission policy: Support full restart, selected-stage restart, resume from returning stage or configured revalidation.

Acceptance: The system identifies which decisions were preserved and which were invalidated.

24. Rejection and Cancellation

Rejection behavior

- Keep source record as Draft
- Allow or prohibit resubmission
- Require a new instance
- Preserve valid documents
- Notify configured parties
- Allow controlled appeal where applicable

Cancellation triggers

- Requester withdrawal
- Source deletion or archive
- Source becomes invalid
- New workflow supersedes old
- Project or relationship closure
- Controlled administrative cancellation

WF-FR-042 - Cancellation confirmation: Cancel pending work items, release locks and confirm source-module outcome.

Acceptance: Cancellation reason, actor and effect are auditable.

25. Record Locking

Lock type	Example
No lock	Advisory review
Submitted fields	Prevent amount, supplier or scope edits
Full record	Freeze a controlled record during final approval
Financial fields	Allow descriptive edits but freeze financial impact
Documents	Freeze controlled revisions referenced by the submission
Status transition	Prevent other transitions while workflow is active
Stage-specific	Apply different locks during legal, finance or executive review

WF-FR-043 - Lock coordination: The engine requests a lock and waits for source confirmation.

Acceptance: A failed lock blocks or routes the workflow according to policy; it never proceeds as if locked.

26. Material Changes and Approval Invalidation

Each workflow version defines material fields, document types, tolerance thresholds and impacted stages. Source modules publish version changes or respond to a revalidation request.

Change category	Examples	Default response
Non-material	Label, typo, internal note	Preserve approval when explicitly allowed
Material value	Amount, price, quantity, currency	Invalidate configured financial stages
Material party	Supplier, client, employee, contract party	Restart affected route
Material scope	Scope, work package, project, unit	Cancel or full restart
Controlled document	Contract terms, drawing or policy revision	Invalidate stages that reviewed the prior revision

WF-FR-044 - Change trace: Show changed fields, before/after values and the invalidation decision.

Acceptance: Approvers and auditors can explain why approval was preserved or restarted.

27. Instance and Work-Item Lifecycles

Instance: Created -> Initializing -> Active -> Waiting for Decision -> Approved -> Source Finalization Pending -> Completed

Alternative instance states include Waiting for Information, Returned, Rejected, Cancelled, Superseded, Suspended, Escalated, Override Applied and Failed.

Work item: Pending -> Assigned -> Opened -> Decision Pending -> Decided -> Completed

Alternative work-item states include Delegated, Escalated, Reassigned, Expired, Cancelled, Superseded and Access Revoked.

WF-FR-045 - State-machine integrity: Persist every transition with expected-version concurrency checks.

Acceptance: Invalid state transitions are rejected and produce a correlation ID for support.

28. Delegation and Substitute Approvers

Delegation field	Requirement
Scope	Module, workflow, company, project, decision type and value ceiling
Period	Effective start and end date with no retroactive authority
Restrictions	Non-delegable categories, confidential records and separation rules
Identity	Preserve delegator and acting user on the decision
Revocation	Immediate for future decisions; prior decisions remain auditable

WF-FR-046 - Delegated authority: A delegate cannot exceed the delegator authority or their own record access.

Acceptance: Resolution excludes any work item outside the delegation scope.

WF-FR-047 - Substitution: Resolve named deputy, position deputy, manager-of-manager or escalation role when the primary is unavailable.

Acceptance: Substitution is instance-specific and does not create permanent role membership.

29. SLA Timers and Escalation

SLA types

- Decision time
- Review time
- Information-response time
- Correction time
- Total workflow duration
- Source-finalization time

Calendars

- Calendar time
- Business hours
- Company calendar
- Country holidays
- Project calendar
- Shift calendar

Escalation actions

- Reminder
- Priority increase
- Notify manager
- Add substitute
- Reassign
- Add stage
- Suspend process
- Route to executive
- Administrative intervention

WF-FR-048 - Durable timers: Persist timers so deployment, restart or worker failover cannot lose them.

Acceptance: Timer state can be reconstructed from stored due dates, calendars and pause events.

WF-FR-049 - SLA pause trace: Record every pause, reason, start and resume.

Acceptance: Displayed elapsed time reconciles with the audit trace.

30. Acknowledgement Workflows

Acknowledgement confirms receipt or understanding and is separate from approval. It applies to policies, HSE instructions, document receipt, contract notices, meeting minutes, emergency notices, payslips and portal transmittals.

WF-FR-050 - Acknowledgement evidence: Record user, organization where applicable, exact document revision or notice version, time and confirmation statement.

Acceptance: A changed revision requires a new acknowledgement when configured.

31. Automated Validations and Source Actions

Validation example	Owning module
Budget available	Finance / Project Cost Control
Contract active	Contracts
Required documents present	Documents
Supplier qualified	Suppliers / Procurement
Timesheet locked	HR and Payroll
Inspection passed	QA/QC
Permit valid	Government and Legal
User active	Identity / HR

Automated validations return Passed, Failed, Warning, Not Available or Retry Required. The engine orchestrates the response; it must not copy the source module business rule.

WF-FR-051 - Source finalization contract: Source action requests include instance, source version, approved decision set, idempotency key and correlation ID.

Acceptance: A repeated request returns the original source result and does not duplicate the business transition.

32. Notifications and Event Centre Integration

- Approval requested
- Work item reassigned or delegated
- Information requested
- Submission returned
- Workflow approved or rejected
- SLA approaching or breached
- Workflow escalated
- Source finalization failed
- Override applied

The Workflow Engine publishes domain events. The Notifications & Event Centre decides channel, timing, quiet hours, grouping, digests, escalation delivery and external-provider behavior. Notification text must use permission-safe projections and minimal sensitive content.

WF-FR-052 - Notification deduplication: Publish stable event IDs and idempotency keys.

Acceptance: Retries do not create duplicate in-app or external notifications.

33. Mobile Approval Experience

- Approval queue
- Source summary
- Key values and changed fields
- Controlled document view
- Workflow timeline
- Approve, reject, return and request information
- Comment and attachment
- Delegation where permitted

Online finalization rule

Mobile may cache source summaries and draft a decision, but formal decision submission requires online server validation of user, device, permission, source version, work-item state and separation of duties.

WF-FR-053 - Stale mobile decision: Preserve a drafted comment when the server rejects a stale decision.

Acceptance: The user sees the new source version and can review before resubmitting.

34. Client and Supplier Portal Participation

- Submit quotation or claim
- Respond to clarification
- Acknowledge Purchase Order
- Submit material proposal
- Respond to NCR
- Confirm client selection
- Accept issued document
- Respond to contract notice

External participants never access internal workflow configuration, internal-only comments, competing submissions or confidential approval history. Portal projections expose only external-facing stage, requested action, permitted comments and published outcome.

WF-FR-054 - External response integrity: Bind each external response to the external account, organization, relationship scope and submission revision.

Acceptance: The engine rejects responses after access expiry or relationship termination.

35. Emergency Override and Controlled Bypass

Emergency override

Use cases include HSE emergencies, urgent legal deadlines, critical site continuity, system outage and emergency procurement. Override requires explicit capability, mandatory reason, evidence, maximum scope, temporary effect, oversight notification and post-event review.

Requested -> Authorized -> Applied -> Temporary Action Completed -> Post-Event Review -> Ratified / Reversed / Corrective Action -> Closed

Controlled bypass

A bypass is a configured exemption proving that a workflow or stage is not applicable, such as low-value preauthorization, legal exemption or duplicate prior approval. It requires a rule, reason, evidence and audit event.

WF-FR-055 - No hidden skip: Do not provide a generic administrator Skip Approval button.

Acceptance: Every override or bypass follows a configured capability and leaves a complete trace.

36. Simulation and Test Environment

Simulation evaluates a synthetic, anonymized, test-tenant or dry-run record without creating live work items or changing source data.

- Selected workflow/version
- Condition results
- Matrix route
- Resolved and excluded participants
- Delegations and conflicts
- Parallel paths and joins
- SLA dates and escalation path
- Completion action
- Errors and warnings

WF-FR-056 - Simulation reproducibility: Store the test payload and configuration versions with the result.

Acceptance: The same historical inputs can be rerun after configuration changes for comparison.

37. Workflow Versioning and Migration

New submissions use the active version. Existing instances remain on their original version unless a controlled migration is performed.

Migration option	Use
Remain on old version	Default and safest path
Migrate selected instances	Only after compatibility validation and node mapping
Cancel and restart	When material routing or source version changes require a new process
Administrative mapping	For urgent correction with preserved decisions and explicit reason

WF-FR-057 - Migration trace: Preserve original version, decisions, mapping, reason and impacted participants.

Acceptance: Auditors can view pre- and post-migration paths.

38. Cross-Module Workflow Registry

Module	Workflow candidates
Projects	Project creation, baseline, budget revision, closure and major status changes
Units and Sales	Reservation, discount, price change, buyer change, variation and handover acceptance
Documents	Review, controlled publication, revision approval, transmittal release and required reading
CRM	Discount, commercial proposal and stage exception
Contracts	Contract review, approval, amendment, renewal, termination and notice issuance
Finance	Journal posting, payment, refund, credit note, budget transfer, close and write-off
Procurement	Requisition, RFQ release, supplier selection, PO, change order and emergency procurement
Inventory	Adjustment, write-off, transfer exception and negative-stock override
Assets	Purchase, disposal, transfer, assignment and maintenance exception
Work Progress	Progress submission, quantity correction and certification
QA/QC and Handover	Inspection acceptance, NCR closure, defect closure and handover readiness
HSE	Incident classification, investigation closure, stop-work release and permit-to-work
Government and Legal	Permit submission, legal response, appeal, fine settlement and legal notice
HR and Payroll	Hiring, offer, salary change, leave, timesheet, payroll, bonus and termination
Reporting	KPI publication, executive report issue and controlled distribution
BIM/3D	Model publication, federated release and client-visible release
Portals	Client variation, supplier submission, claim, invoice, defect closure and acknowledgement

39. Domain and Data Model

Core entities		
WorkflowDefinition	WorkflowVersion	WorkflowNode
WorkflowTransition	WorkflowCondition	WorkflowTrigger
WorkflowSelectionRule	ApprovalMatrix	ApprovalMatrixRule
WorkflowInstance	WorkflowStageInstance	WorkflowNodeInstance
WorkflowWorkItem	WorkflowParticipant	ParticipantResolution
WorkflowDecision	DecisionCondition	InformationRequest
WorkflowReturn	WorkflowLock	WorkflowSLA
WorkflowTimer	WorkflowEscalation	WorkflowDelegation
WorkflowSubstitute	WorkflowOverride	WorkflowBypass
WorkflowCancellation	WorkflowMigration	SourceActionRequest
SourceActionResult	WorkflowAttachment	WorkflowComment
WorkflowAuditEvent	WorkflowMetricSnapshot	

Workflow instance core fields

id, tenant_id, company_id, project_id, workflow_definition_id, workflow_version_id, source_module, source_entity_type, source_entity_id, source_record_version, trigger_action, submitted_by, submitted_at, current_stage, current_node, status, priority, confidentiality, sla_due_at,

completed_at, correlation_id, record_version

40. Runtime and State-Machine Architecture

Source Submission -> Transactional Event -> Workflow Selection -> Instance Creation -> Participant Resolution -> Work Item Creation -> Decision Processing -> Transition Evaluation -> Source Finalization -> Completion

- Transactional outbox
- Idempotent workflow start
- Optimistic concurrency
- Durable state machine
- Durable timers
- Retry queues
- Dead-letter handling
- Correlation IDs
- Compensating actions
- Complete execution trace

WF-FR-058 - Atomic transitions: Commit work-item decisions and state transitions atomically within the workflow runtime.

Acceptance: A crash cannot record a decision without its resulting state or vice versa.

WF-FR-059 - At-least-once event safety: Consumers must be idempotent and tolerate repeated or out-of-order events.

Acceptance: Duplicate source events do not create duplicate instances or transitions.

41. Domain Events and Integration Contracts

Consumed events	Published events
RecordSubmittedForApproval	WorkflowStarted
RecordVersionChanged	WorkflowWorkItemCreated
RecordCancelled	WorkflowDecisionRecorded
UserDeactivated	WorkflowInformationRequested
UserAssignmentChanged	WorkflowReturned
DelegationCreated	WorkflowApproved / WorkflowRejected
ProjectClosed	WorkflowEscalated / WorkflowOverridden
CompanyConfigurationChanged	WorkflowSourceFinalizationRequested
SourceActionCompleted / Failed	WorkflowCompleted / WorkflowFailed

WF-FR-060 - Event schema versioning: Every event includes event name, version, tenant, source, correlation and idempotency metadata.

Acceptance: Unknown incompatible versions are rejected to a dead-letter queue with no partial processing.

42. API Contract

Decision APIs

```
GET /workflow/work-items
GET /workflow/work-items/{id}
POST /workflow/work-items/{id}/approve
POST /workflow/work-items/{id}/reject
POST /workflow/work-items/{id}/return
POST /workflow/work-items/{id}/request-information
POST /workflow/work-items/{id}/acknowledge
POST /workflow/work-items/{id}/delegate
```

Instance and definition APIs

```
POST /workflow/instances
GET /workflow/instances/{id}
POST /workflow/instances/{id}/cancel
POST /workflow/instances/{id}/resubmit
GET /workflow/instances/{id}/history
POST /workflow/definitions
POST /workflow/definitions/{id}/versions
```

```
POST /workflow/versions/{id}/validate
POST /workflow/versions/{id}/simulate
POST /workflow/versions/{id}/publish
```

All write APIs require an idempotency key, expected workflow/work-item version, expected source-record version, user and device context, authorization capability and audit metadata. Error responses include a stable code and correlation ID.

43. Frontend Architecture

Component		
MyApprovalsPage	ApprovalQueue	ApprovalCard
ApprovalDetailPage	SourceRecordSummary	DecisionPanel
DecisionCommentEditor	DecisionAttachmentPanel	WorkflowTimeline
WorkflowStageProgress	DelegationPanel	InformationRequestPanel
ReturnForCorrectionModal	WorkflowHistoryDrawer	WorkflowDashboard
WorkflowDefinitionList	WorkflowBuilder	WorkflowCanvas
WorkflowNodeLibrary	WorkflowPropertiesPanel	ApprovalMatrixEditor
ConditionBuilder	ParticipantRuleEditor	SLAPolicyEditor
EscalationEditor	WorkflowSimulationPanel	WorkflowVersionCompare
WorkflowFailureConsole	OverrideReviewPage	

- Server-state cache with URL-addressable filters
- Optimistic UI only for reversible non-decision interactions
- No optimistic final decision
- Permission-invalidated caches
- Virtualized large queues
- Accessible keyboard and mobile interactions
- Independent failure boundaries for dashboard widgets

44. Security, Reauthentication and Signatures

- Tenant, company, project and record isolation
- Field-level redaction
- Decision capability separate from view/edit permissions
- Encrypted decision evidence and attachments
- No confidential values in notification payloads
- Sensitive read audit
- Rate limiting and anti-replay controls
- Secure session and device context

Critical-decision confirmation

- Password re-entry
- MFA confirmation
- Biometric confirmation on trusted mobile device
- Confirmation statement
- Reason code
- Future qualified e-signature or certificate integration

Signature distinction

Authentication confirmation, platform acknowledgement, internal digital approval and legally qualified electronic signature are different capabilities and must never be labeled as equivalent by default.

45. Permissions and Confidentiality

Confidentiality class	Examples
Standard	Routine project and task approvals
Internal	Internal operational decisions
Financial Restricted	Payments, budgets and financial postings
Payroll Restricted	Payroll runs, payslips and compensation changes

Confidentiality class	Examples
HR Confidential	Hiring, discipline, termination and sensitive employment cases
Legal Restricted	Legal opinions, litigation and regulatory responses
Executive	High-value, strategic or ownership decisions
HSE Critical	Emergency, stop-work and incident decisions
External Party Restricted	Portal submissions and external communications

WF-FR-061 - Minimum decision projection: An approver may receive only the fields needed to decide, even when they cannot open the full source record.

Acceptance: The projection is generated by the source module and version-bound.

WF-FR-062 - No admin authority inheritance: Workflow or platform administration capabilities do not imply approval authority.

Acceptance: An administrator without a business role cannot decide live work items.

46. Audit, Compliance and Retention

- Workflow selected and version used
- Conditions evaluated
- Participants resolved and excluded
- Work item assigned, opened and reassigned
- Decision and evidence
- Information request and response
- Return and resubmission
- Delegation and escalation
- SLA pause/resume
- Record version changes and invalidation
- Cancellation, migration, override and bypass
- Source action request and result
- Completion or failure

Audit events include user, acting user, tenant, company, project, source record, workflow version, timestamp, device/session, correlation ID, before/after state and reason. Retention may differ by workflow category and must support legal hold.

WF-FR-063 - Audit reconstruction: Provide one chronological execution view that does not depend on application server logs.

Acceptance: An auditor can identify why each participant was selected and how the final state was reached.

47. Search, Reporting and Workflow Analytics

Search

Search indexes workflow number, source reference, requester, approver, company, project, stage, status and safe source title. Confidential source data is not indexed unless explicitly allowed.

Analytics

- Average workflow and stage duration
- Approval, rejection and return rates
- Information-request rate
- SLA compliance
- Escalation and delegation rates
- Override and bypass rates
- Participant-resolution failures
- Source-finalization failures
- Bottlenecks by module, company, project and workflow

Analytics guardrail

Workflow speed, queue age and decision patterns must not automatically become employee-performance ratings. Any HR use requires separate policy, context and authorization.

48. Administration and Platform Configuration

- Workflow categories and definitions
- Approval matrices
- Participant resolvers

- Decision options
- Comment and attachment requirements
- SLA calendars
- Escalation policies
- Delegation and substitute rules
- Separation-of-duty rules
- Material-change fields
- Lock policies
- Rejection, return and cancellation behavior
- Override and bypass policies
- Source actions
- Notification hooks
- Mobile decision rules
- Portal participation
- Retention and confidentiality
- Custom fields

WF-FR-064 - Configuration versioning: Effective configuration used by selection, resolution, SLA and decision validation must be versioned or snapshotted.

Acceptance: Historical instances remain explainable after configuration changes.

49. Non-Functional Requirements

Category	Requirement
Performance	Queue P95 under 2 seconds; decision processing P95 under 3 seconds excluding source finalization; selection under 1 second.
Availability	Durable state and timers; no decision loss during service failure; retry-safe source actions.
Scalability	Thousands of definitions, high-volume invoice/payroll peaks and long-running legal workflows.
Security	Deny-by-default, tenant isolation, least privilege, reauthentication and encrypted evidence.
Accessibility	WCAG-aligned keyboard, screen-reader, contrast, focus and non-color status support.
Localization	Time zones, business calendars, languages, currencies and locale-aware dates.
Maintainability	Versioned schemas, observability, automated tests and backward-compatible contracts.
Data integrity	Immutable decisions, optimistic concurrency and idempotent commands/events.

50. Concurrency, Resilience and Error Handling

- Two approvers decide simultaneously
- Approver is deactivated while item is open
- Source record changes during approval
- Source module is unavailable after approval
- Source action succeeds but confirmation is delayed
- Duplicate trigger or decision request
- Timer fires during deployment
- External portal relationship expires during response

Error class	Required behavior
Validation	Preserve user draft and show actionable field/record reason
Permission	Reject action, redact details and audit the attempt
Version conflict	Show current version and require review before retry
Participant resolution	Pause in intervention state and follow fallback
Source unavailable	Remain pending, retry safely and notify operations
Permanent source rejection	Expose source validation error and route to configured correction or administrator
Duplicate	Return original result

Error class	Required behavior
Unexpected runtime	Fail safely, preserve state and emit a correlation ID

51. Observability and Operational Support

Technical metrics

- Workflow starts and failures
- Selection latency
- Participant-resolution failures
- Decision latency and conflicts
- Timer backlog and drift
- Source-action retries
- Dead-letter volume
- Queue age
- API error rate
- Database and broker health

Operational tools

- Workflow instance trace
- Failed workflow console
- Retry source action
- Replay safe event
- Resolve participant manually with authority
- Suspend/resume instance
- Compare definition versions
- Export audit package

WF-FR-065 - Safe intervention: Operational support actions must be capability-controlled, reasoned and audited.

Acceptance: Support users cannot modify finalized decisions or silently jump workflow state.

52. Quality Assurance and Test Strategy

Testing must include unit, state-machine, contract, integration, security, concurrency, performance, migration, accessibility and end-to-end scenarios. Every published workflow template requires a validated simulation suite.

- Workflow selection and matrix thresholds
- Currency conversion and historical rate
- Sequential and parallel routing
- Quorum and simultaneous decisions
- Rejection, return and information requests
- Delegation, substitute and separation of duties
- SLA calendars, holidays and daylight-saving changes
- Locking and material changes
- Source finalization and delayed events
- Duplicate triggers and decisions
- Migration, override and bypass
- Multi-company, project and portal isolation
- Mobile approval and stale versions
- High-volume invoice and payroll peaks
- Failure recovery and dead-letter replay

Critical scenarios

- Purchase Order amount changes after Finance approval.
- Approver leaves the company while work items are active.
- Requester is also the department manager.
- Two parallel approvers decide simultaneously.
- Source event is delivered three times.
- Contract document revision changes after legal review.
- Payroll is approved but Finance rejects posting.
- Delegation exceeds the permitted value.
- Workflow version changes while an instance is active.
- Emergency override occurs during a service outage.
- External supplier resubmits after internal rejection.
- Workflow administrator attempts to approve without business authority.

53. Acceptance Criteria

- AC-01 - Acceptance criterion:** A source module can start a workflow with a versioned idempotent trigger and receive the same instance on retry.
- AC-02 - Acceptance criterion:** Selection trace identifies the exact rule, configuration version and workflow version.
- AC-03 - Acceptance criterion:** Sequential and parallel templates pass simulation and live end-to-end tests.
- AC-04 - Acceptance criterion:** A user cannot approve a work item without explicit decision capability and record context.
- AC-05 - Acceptance criterion:** Self-approval and configured duty conflicts are blocked or rerouted.
- AC-06 - Acceptance criterion:** Every decision records user, acting user, source version, workflow version, time, result and required evidence.
- AC-07 - Acceptance criterion:** Material source changes invalidate the configured stages and display the reason.
- AC-08 - Acceptance criterion:** Durable SLA timers survive service restart and respect business calendar pauses.
- AC-09 - Acceptance criterion:** Delegation remains within time, scope, value and confidentiality limits.
- AC-10 - Acceptance criterion:** Approved workflows remain pending until the source module confirms finalization.
- AC-11 - Acceptance criterion:** Mobile final decisions require server validation and never show success prematurely.
- AC-12 - Acceptance criterion:** Portal participants cannot access internal workflow history or unrelated records.
- AC-13 - Acceptance criterion:** Emergency override preserves skipped stages and creates mandatory review.
- AC-14 - Acceptance criterion:** Audit view reconstructs selection, routing, decisions, changes and finalization.
- AC-15 - Acceptance criterion:** The engine meets defined performance, security, accessibility and isolation targets.

54. Delivery Phases and Rollout

Phase	Scope
Phase 1 - Approval foundation	Definitions, versions, sequential routing, work-item queue, decisions, locks, notifications, audit and basic matrices.
Phase 2 - Enterprise routing	Parallel paths, conditions, dynamic participants, delegation, substitutes, separation of duties, SLA, escalation, return and information requests.
Phase 3 - Advanced governance	Quorum, conditional approval, simulation, migration, emergency override, controlled bypass, portal stages, mobile approval and analytics.
Phase 4 - Platform-wide adoption	Integrate source actions and templates across all ConstructionOS modules; retire module-specific approval code.

Rollout should begin with high-value, well-defined processes such as Purchase Order approval, contract approval and payroll run approval. Each source-module integration must pass contract, security and recovery tests before production activation.

55. Migration and Adoption Strategy

1. Inventory every existing approval and review point across modules.
2. Classify each as approval, review, verification, acknowledgement or source-owned validation.
3. Define standard decision and source-action contracts.
4. Build reusable template families and matrices.
5. Run old and new routing in shadow mode for selected processes.
6. Migrate only new submissions first; preserve active legacy workflows until completion.
7. Enable operational dashboards and failure support before expanding volume.
8. Decommission hard-coded module approval paths after acceptance and audit sign-off.

WF-FR-066 - Legacy trace preservation: Historical module decisions remain accessible after migration.

Acceptance: The platform links legacy history without rewriting or falsely reclassifying it as engine-native decisions.

56. Risks and Open Decisions

Risk / decision	Mitigation or required decision
Over-configurable workflows become difficult to govern	Template library, ownership, validation, simulation and publication rights.
Source modules expose inconsistent version or action contracts	Create a platform integration standard and certification test suite.
Approver resolution depends on incomplete organization data	Unresolved-participant intervention state and data-quality dashboards.

Risk / decision	Mitigation or required decision
Emergency override is abused	Narrow capabilities, value/scope limits, oversight alerts and mandatory review.
Long-running workflows break after organization changes	Effective-dated participant resolution and substitute policy.
Workflow changes invalidate historical reproducibility	Immutable versions and configuration snapshots.
Approval fatigue and bottlenecks	Threshold matrices, advisory review, digests and analytics; never auto-approve hidden exceptions.
Legal meaning of digital approval varies by jurisdiction	Clearly label internal approval; add qualified e-signature integrations only after legal design.
Need for BPMN compatibility	Open decision: internal node model only or future BPMN import/export adapter.
Multi-region data residency	Open decision for deployment topology before regulated-market rollout.

57. Definition of Done

- All acceptance criteria pass in automated and manual tests.
- At least three production-grade source-module integrations are complete.
- Workflow builder, simulation, publication and rollback paths are documented.
- Security review confirms deny-by-default and separation-of-duty enforcement.
- Load tests cover expected financial and payroll peaks.
- Durable timer and failure-recovery tests pass.
- Mobile and portal decision projections pass accessibility and security review.
- Audit package is accepted by compliance stakeholders.
- Operational dashboards, alerts, runbooks and support tools are available.
- No known critical or high-severity defects remain open.
- Platform Configuration and permission matrices are documented and versioned.
- API and event contracts are registered in the platform contract registry.

58. Page and Flow Registry

Surface / flow	Primary users	Key actions
My Approvals	Approvers	Filter, open, decide, request information, delegate
My Submissions	Requesters	Track, answer, withdraw, correct, resubmit
Team Approval Queue	Managers	Monitor workload, overdue and unresolved items
Approval Detail	Approvers / reviewers	Review exact version, documents, history and decide
Workflow Dashboard	Managers / administrators	Monitor bottlenecks, failures, SLA and overrides
Workflow Definitions	Workflow administrators	Create, version, validate and publish
Workflow Builder	Workflow administrators	Design nodes, transitions, conditions and actions
Approval Matrix Editor	Authorized administrators	Manage thresholds, roles, values and effective dates
Delegations	Users / managers	Create, review, revoke and audit bounded delegation
Simulation	Workflow administrators / QA	Run test payloads and inspect routing trace
Workflow Instances	Support / auditors	Inspect runtime state, history and integration results
Failed Workflows	Operations	Retry safe actions and resolve controlled failures
Override Review	Oversight roles	Review, ratify, reverse or require corrective action
Mobile Approval	Approvers	Review and submit online-confirmed decisions
Portal Workflow Projection	Clients / suppliers	Submit, respond, acknowledge and view external status

Final architectural position: The engine is the versioned, permission-aware and auditable decision-orchestration infrastructure for every controlled business transition across ConstructionOS.